

Week 3. Homework 1. SQL Basic

Due date: **September 12, 11:59 pm**

Submission Guidelines

- **Dataset:** You must use the provided *IMDB.sqlite* dataset to answer all questions.
- **Code and Output:** Copy your code and attach screenshots of your outputs under each question.
- **Submission Format:** Each student is required to push one Word or PDF file to a designated GitHub repository you create specifically for this week's assignment.
- **Repository Name:** Use the format *Week 3-HW1-LastName*, where *LastName* is your last name.
- **Code Quality:** Ensure your code is organized for easy understanding.
- **AI Usage:** You should make a serious attempt at each problem prior to using an LLM. Remember, you may be tested on SQL knowledge without assistance in a technical interview. If prompting an LLM, I suggest telling it your starting point, your thought process, and asking it to guide you. I've done this learning SQL and it is helpful and I learned more that way. Disclose your usage at the end of the file; you will not be penalized regardless, but this give me some information.

Questions

1. Directors and Their Departments

List the names of all directors along with their respective departments.

	name	department
1	James Cameron	Directing
2	Gore Verbinski	Directing
3	Sam Mendes	Directing
4	Christopher Nolan	Directing
5	Andrew Stanton	Directing
6	Sam Raimi	Directing
7	Byron Howard	Directing
8	Joss Whedon	Directing
9	David Yates	Directing
10	Zack Snyder	Directing

```
-- Question 1
SELECT name, department
FROM directors
```

2. Popular Movies

Find the titles of all movies with a popularity score greater than 5.

<pre>--Question 2 SELECT title FROM movies WHERE popularity > 5</pre>	1	Avatar
	2	Pirates of the Caribbean: At World's End
	3	Spectre
	4	The Dark Knight Rises
	5	John Carter
	6	Spider-Man 3
	7	Tangled
	8	Avengers: Age of Ultron
	9	Harry Potter and the Half-Blood Prince
	10	Batman v Superman: Dawn of Justice

3. Directors in the Directing Department

Retrieve the names and IDs of directors who are in the 'Directing' department and have a gender value of '2'.

<pre>--Question 3 SELECT name, id FROM directors WHERE department LIKE '%Directing%' AND gender LIKE '2'</pre>		name	id
	1	James Cameron	4762
	2	Gore Verbinski	4763
	3	Sam Mendes	4764
	4	Christopher Nolan	4765
	5	Andrew Stanton	4766
	6	Sam Raimi	4767
	7	Byron Howard	4768
	8	Joss Whedon	4769
	9	David Yates	4770
	10	Zack Snyder	4771

4. Directors with 'Sam' in Their Name

Find all directors whose names include the word 'Sam'.

	name
1	Sam Mendes
2	Sam Raimi
3	Sam Fell
4	Sam Weisman
5	Karyn Kusama
6	Sam Taylor-Johnson
7	Sam Miller
8	Sammo Hung
9	Sam Peckinpah
10	Sam Levinson
11	Sam Firstenberg


```

--Question 4
SELECT name
FROM directors
WHERE name LIKE '%Sam%'

```

5. First 10 Movies by Release Date

Show the first 10 movies in the database, sorted by release date in ascending order.

	title
1	Intolerance
2	The Big Parade
3	Metropolis
4	Pandora's Box
5	The Broadway Melody
6	Hell's Angels
7	A Farewell to Arms
8	42nd Street
9	She Done Him Wrong
10	It Happened One Night


```

--Question 5
SELECT title
FROM movies
ORDER BY release_date ASC
LIMIT 10

```

6. 'Star' Movies with High Ratings

List all movies with the word 'Star' in the title and a vote average greater than 7.

	title
1	Star Trek Into Darkness
2	Star Trek
3	Star Wars: Episode III - Revenge of the Sith
4	Stardust
5	Star Trek II: The Wrath of Khan
6	The Fault in Our Stars
7	Star Wars
8	Star Wars: Clone Wars: Volume 1
9	20 Feet from Stardom

--Question 6

```
SELECT title
FROM movies
WHERE title LIKE '%Star%'
AND vote_average > 7
```

7. Movies with Limited Revenue (1995–2000)

Find the titles of movies released between January 1, 1995, and December 31, 2000, with revenues not exceeding 10 million.

	title
1	Volcano
2	Déjà Vu
3	Mighty Joe Young
4	Little Nicky
5	Inspector Gadget
6	Anna and the King
7	Primary Colors
8	Proof of Life
9	Lucky Numbers
10	The Family Man
11	Up Close & Personal
12	Mercury Rising

--Question 7

```
SELECT title
FROM movies
WHERE (release_date BETWEEN '1995-01-01'
AND '2000-12-31')
AND revenue <= 10000000
```

8. Directors with Specific 'uid' Values

List the names of directors who have a uid greater than 7,000 or less than 100.

	name
1	Sam Mendes
2	Andrew Stanton
3	Sam Raimi
4	Byron Howard
5	Joss Whedon
6	David Yates
7	Zack Snyder
8	Bryan Singer
9	Marc Forster
10	Rob Marshall
11	Marc Webb
12	Anthony Russo

```
--Question 8
SELECT name
FROM directors
WHERE uid > 7000 OR uid < 100
```

9. Movies with Specific Vote Counts

Retrieve the titles of movies that have:

- A vote counts between 50 and 100, inclusive.
- A vote average of exactly 7.0.

	title
1	Angela's Ashes
2	A Better Life
3	Red Riding: In the Year of Our Lord 1974
4	Quo Vadis
5	Another Year
6	Salvador
7	The Black Stallion
8	Elling
9	Fireproof

```
--Question 9
SELECT title
FROM movies
WHERE vote_count BETWEEN '50' AND '100'
AND vote_average LIKE '7.0'
```

10. Directors Named 'John' with Specific 'uid' Values

Find the names of directors who:

- Have 'John' in their name.
- Have a uid between 5,000 and 10,000 (exclusive).

```
--Question 10
SELECT name, uid
FROM directors
WHERE name LIKE '%John%'
AND (uid > '5000' AND uid < '10000')
```

	name	uid
1	John Lasseter	7879
2	John Milius	8328
3	John Singleton	6482
4	John Guillermin	6778
5	John McNaughton	9200
6	John Ottman	9039
7	John Huston	6593
8	John Ford	8500

11. Movies with Odd 'uid' and Even Vote Count

Retrieve the titles of movies that meet the following criteria:

- The uid is an odd number.
- The vote count is an even number and greater than 500.

```
--Question 11
SELECT title
FROM movies
WHERE uid % 2 = 1
AND (vote_count % 2 = 0 AND vote_count > 500)
```

	title
1	Avatar
2	Pirates of the Caribbean: At World's End
3	Spectre
4	John Carter
5	Spider-Man 3
6	Tangled
7	Pirates of the Caribbean: On Stranger Tides
8	The Hobbit: The Battle of the Five Armies
9	Titanic
10	Battleship
11	Jurassic World
12	Iron Man 3

12. Movies with Specific Ratings or Budgets

Find the titles of movies that meet any of the following criteria:

- Have a vote average between 6.0 and 8.0, with a vote count of exactly 200.
- OR have a budget greater than 100,000 but less than 500,000.

	title
1	Halloween
2	Control
3	Snow Flower and the Secret Fan
4	Night of the Living Dead
5	Sleeper
6	Open Water
7	Conversations with Other Women
8	42nd Street
9	Metropolitan
10	Napoleon Dynamite
11	Monty Python and the Holy Grail
12	Intolerance

--Question 12

SELECT title

FROM movies

WHERE (vote_average **BETWEEN** '6.0' **AND** '8.0' **AND** vote_count **LIKE** '200')

OR (budget > '100000' **AND** budget < '500000')